Twin Cities Performing Arts Database

Introduction to Database Management Systems Project

Tu Tran, Yosephine Manihuruk, Natalia Morales Flores

Motivation and Purpose

To create a comprehensive calendar of performances with the ability to browse by features, historical record of performances, and to potentially become a shared database of artists and crew to connect with opportunities.

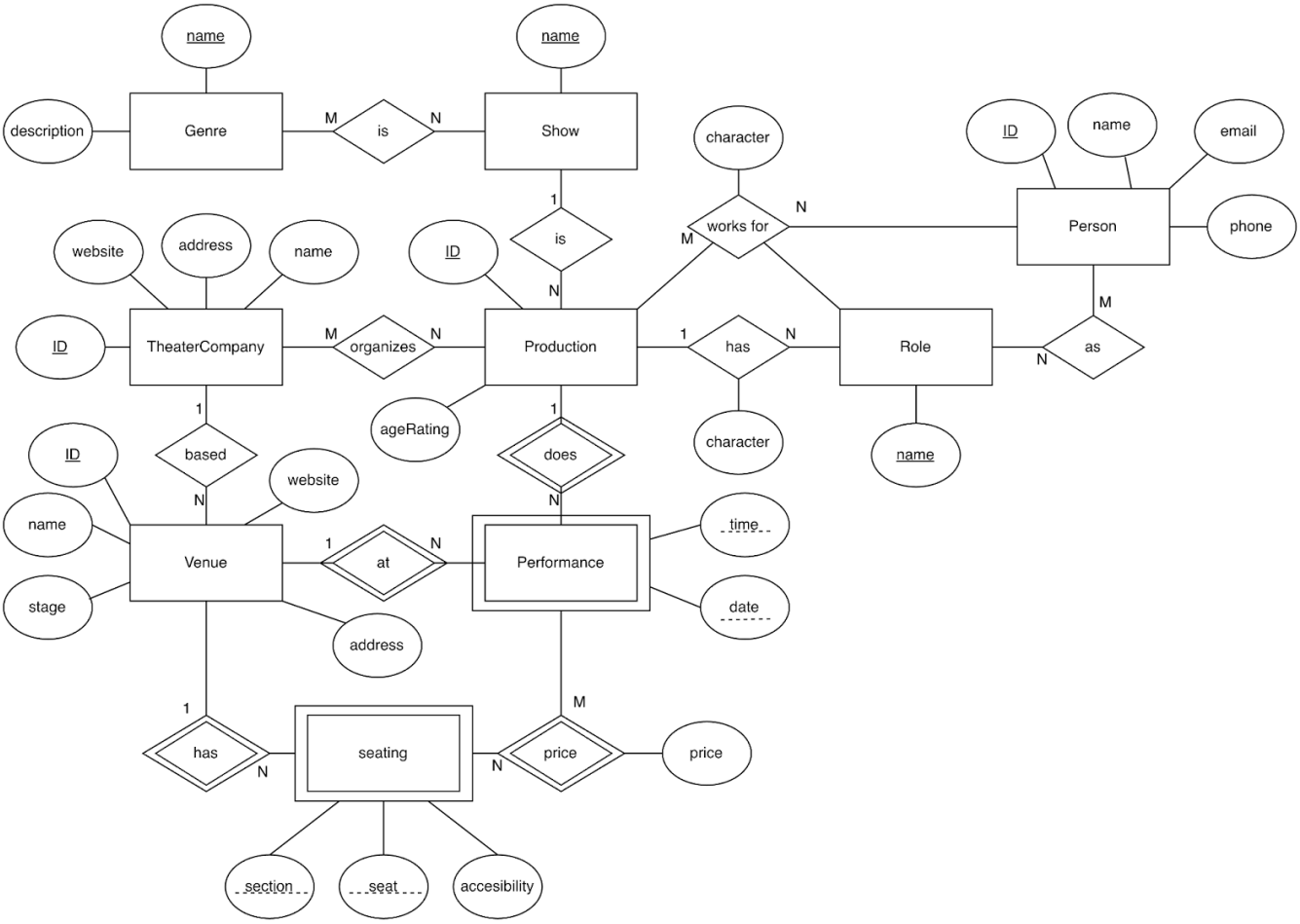
Sample queries

a. Find all shows playing during a specific date range
b. Find all shows an artists worked in
c. Find all past / upcoming shows at a specific venue
d. Find all artists worked in a specific show
e. Find all shows by genre

Meeting log

Date	Length (mins)	Roles	Notes
Thu Feb 19	90	User: Natalia Data modelers: Tu, Yosephine	Define entities, attributes and relationships. Create initial ERD.
Tue Feb 24	90	User: Yosephine Data modelers: Tu, Natalia	Complete initial ERD diagram and create sample data. Identified Performance as a weak entity. Connected TheaterCompany and Venue since various seem to be the same
Mon Mar 2	90	User: Tu Data modelers: Natalia, Yosephine	Refine ERD diagram: identified derived data and created a new entity based on recommendations through feedback. Finished report. Prepare for presentation.

Data model: ERD



Documentation

- Show Entity
    - The work/title (the thing being produced) independent of any specific theater company or run (e.g. Hamlet)
    - Primary Key (PK): name of the show
    - A show could be multiple genres. A show could be associated with multiple productions.
- Genre Entity
    - A category to label the type of show
    - PK: name of the genre
    - A genre could be associated with multiple shows
- TheaterCompany Entity

    - An organization that stages productions (e.g community theater, private theater, school theater, etc.)
    - PK: ID of Theater company
    - A theater company could organize multiple productions. A theater company could be based in multiple venues.
- Production Entity
    - A theater company's production of a show
    - PK: an ID assigned to each production
    - A production could be organized from one or a collaboration of multiple theater companies. A production is associated with one show. A production could do multiple performances at different venues, date and time. A production has multiple roles.
- Person Entity
    - Any individual who has worked in a production or offers a service that could be used in a production
    - PK: ID of the individual
    - A person could work for multiple productions under different roles.
- Role Entity
    - The role that a person could take on, such as actor, director, crew member, etc.
    - PK: name of the role
    - A role could be associated with multiple persons and multiple productions.
- Performance Weak Entity
    - A single scheduled occurrence of a specific production of a show (e.g., March 10 at 7:30pm)
    - PK: a combination of venue ID, production ID, the date and time of the performance
    - Its parent entities are Production and Venue. A production can have many performances. And each venue can host many performances. But each performance has to belong to a specific production and venue.
- Venue Entity
    - The physical place where a certain production is presented at a certain time
    - PK: a unique ID assigned to each venues
    - A venue could be associated with one theater company and have multiple performances.

- Seating Weak Entity
    - The section and seat in a certain venue for a certain performance
    - PK: a combination of venue ID, section, and seat number
    - Its parent entity is Venue
    - Seating is associated with Venue through a parent relationship and with Seating, creating Price
- Price relationship
    - Based on the Performance and the Seating, the price is assigned
    - An attribute of price was added to specify the amount

Societal Impacts

Potential Benefits

1. Early discovery of local venues: A calendar like this would allow for people to quickly find events, venues, companies, etc. lowering the effort of engaging with the arts in the Twin Cities.
2. Better audience planning: Centralizing basics such as venue and seating capacity can help audiences to plan realistically.
3. Historical record and cultural preservation: Storing past productions creates an archive of what was performed, allowing to notice patterns, gaps, and to contribute to the community memory.
4. Network building and opportunities: A shared database of people and productions can surface connections (who collaborated before, where a designer has worked), potentially supporting hiring, guidance, and collaboration opportunities.

Potential Risks

1. Unequal representation and visibility bias: If the database starts with (or is more complete for) large, professional companies and venues, smaller community groups may be undercounted, making the “default view” of theater less diverse than reality.
2. Privacy and safety risks for individuals: A shared artist/crew database can unintentionally expose personal info (emails, phone numbers, locations, affiliations, work history), and shared pictures could skip asking for consent. In the current context, some cultural performances might attract the attention of biased “surveillance groups”.

3. Copyright and usage: Show descriptions, photos, and promotional text may be copyrighted; storing or redistributing them without clear permissions can create legal and ethical issues.

Costs and Responsibility

1. Asking for consent: make email/phone opt in. Allow “contact via company” instead of personal contact informations
2. Access control: separate views for the general public and admins/professionals.
3. Copyright costs to avoid legal issues